# Hieu T. Le

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## Education

**Texas A&M University** – PhD in Electrical Engineering. Advisor: Sabit Ekin — Jan 2022 - Dec 2026
**Texas A&M University** – MS in Electrical Engineering. — August 2021 - May 2025

## Experience

**Machine Learning Intern, Rider Pricing and Incentives,** Uber – CA — June 2026 – August 2026

- Trained, evaluated, and shipped launch-ready, continuously monitored machine learning pricing models across new markets, lifting global booked trips by 3 million.
- Built a closed-loop multi-agent AutoResearch LLM system automating ML workflows, from ideation to distributed Ray training, providing a scalable, model-agnostic platform that reduced experiment cycle times to 1 hour.
- Enforced production-grade reliability by isolating implementer agents and deploying an independent verification agent, intercepting 95% of faulty model changes before they consumed GPU resources.

**Engineering Intern,** Qualcomm – San Diego, CA — Sep 2025 – Dec 2025

- Engineered an ML-driven resource allocation system utilizing Vision-Transformer-based models for near real-time Visual Quality Assessment (VQA), successfully reducing wireless bandwidth consumption by 20% at equivalent perceived quality.
- Architected a multi-threaded GStreamer-WebRTC video streaming to enforce mobile-edge constraints, executing 180 ms background C++ LibTorch ML inference to drive instantaneous server-side adaptive bitrate control without interrupting real-time streaming.
- Deployed an asynchronous, end-to-end data training pipeline (ingest, preprocess, stream) to teach Machine Learning models to predict rate-distortion curves, achieving a strong correlation ($R^2 = 0.89$) between predicted and actual video quality-bitrate slopes.

**Research Assistant,** Texas A&M University – College Station, TX — September 2021 – Present

- Implemented a hierarchical multi-agent RL framework that optimizes physical hardware under real-world constraints, achieving 8$\times$ signal enhancement and outperforming MAPPO baselines.
- Led a cross-functional team integrating software and hardware for a reconfigurable testbed, from prototype to experimental validation.
- Optimized access-point placement by combining particle swarm / genetic search with gradient descent through Sionna's differentiable ray tracer, improving throughput by 15% over gradient descent alone.

**Research Assistant,** High Performance Computing Center - Texas A&M University — Jan 2024 – June 2024

- Built execution pipelines to benchmark the performance and memory-bound bottlenecks of BERT and ResNet models across fundamentally different silicon architectures (distributed NVIDIA GPUs vs. Graphcore IPUs).

## Projects

**LLM-Driven Dynamic Pricing Optimization via GRPO**

- Optimized the VRAM footprint of a Gemma 4 E4B model to fit multi-generation RL training rollouts on a single GPU by deploying 4-bit QLoRA and gradient checkpointing.
- Fine-tuned the model using Group Relative Policy Optimization (GRPO) to act as an intelligent dynamic pricing agent within a custom, tensorized PyTorch supply-demand environment.

## Skills

**Languages & Frameworks:** Python, C++, C, SQL, Pandas, PyTorch, Ray, vLLM.
**LLMs & Agentic Systems**: LLM fine-tuning (QLoRA/LoRA), multi-agent orchestration, and agentic workflows.